

Autodesk®
Scaleform®

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Autodesk® Scaleform® 4.3

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1 소개

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2 문서 개요

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3 설치와 빌드 의존성

3.1 설치

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3rdParty

Projects

Apps\Samples

Apps\Tutorial

Bin

주:

Include

Lib

Resources

Doc

3.2 데모 컴파일

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3.3 Scaleform Player 에서 SWF 재생 벤치마크

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그림 1:

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3.4 샘플 베이스 컴파일



그림 2:

3.5 Scaleform 빌드 의존성



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주:

3.5.1 AS3_Global.h 파일과 ObjWAS3_Obj_Global.xxx 파일

\ 서로 연관성이 전혀 없는

4 게임 엔진 통합

4.1 플래시 렌더링



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1 단계: 헤더 파일 추가

```
#include "GFx_Kernel.h"  
#include "GFx.h"  
#include "GFx_Renderer_D3D9.h"
```

```
    ne GFx::Loader per application  
Loader          gfxLoader;  
  
    // One GFx::MovieDef per SWF/GFx file  
Ptr<MovieDef>    pUIMovieDef;  
  
    // One GFx::Movie per playing instance of movie  
Ptr<Movie>       pUIMovie;  
  
    // Renderer data  
Ptr<Render::D3D9::HAL> pRenderHAL;  
Ptr<Render::Renderer2D> pRenderer;  
MovieDisplayHandle    hMovieDisplay;
```

2 단계: GFx::System 초기화

```
    // One GFx::System per application  
GFx::System      gfxInit;
```


3 단계: 로더 및 렌더러 생성

```
gfx = new GFxTutorial();  
assert(gfx != NULL);  
if(!gfx->InitGFx())  
    assert(0);
```

```
// Initialize logging -- Scaleform will print errors to the log  
// stream.  
gfxLoader->SetLog(Ptr<Log>(*new Log()));
```

```
// Give the loader the default file opener
```

```
Ptr<FileOpener> pfileOpener = *new FileOpener;  
gfxLoader->SetFileOpener(pfileOpener);
```

```
pRenderHAL = *new Render::D3D9::HAL();  
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr())))  
    return false;
```

4 단계: 플래시 무비 로딩

```
// Load the movie  
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,  
                                     Loader::LoadKeepBindData |  
                                     Loader::LoadWaitFrame1, 0);
```

5 단계: 무비 인스턴스 생성

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);  
assert(pUIMovie.GetPtr() != NULL);
```

```
// Advance the movie to the first frame  
pUIMovie->Advance(0.0f, 0, true);  
  
// Note the time to determine the amount of time elapsed between  
// this frame and the next  
MovieLastTime = timeGetTime();
```

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

6 단계: 디바이스 초기화

```
pRenderHAL->InitHAL(  
    Render::D3D9::HALInitParams(pd3dDevice, presentParams,  
                                Render::D3D9::HALConfig_NoSceneCalls));
```

7 단계: 로스트 디바이스

```
pRendererHAL->ShutdownHAL();
```

8 단계: 자원 할당 및 청소(cleanup)

```
delete gfx;  
gfx = NULL;
```

9 단계: 뷰포트 설정

```
// Use the window client rect size as the viewport.  
RECT windowRect = DXUTGetWindowClientRect();  
DWORD windowWidth = windowRect.right - windowRect.left;  
DWORD windowHeight = windowRect.bottom - windowRect.top;  
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,  
                        windowHeight, windowHeight);
```

10 단계: DirectX 썬으로 렌더링

```
void AdvanceAndRender(void)
{
    DWORD mtime = timeGetTime();
    float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
    MovieLastTime = mtime;

    pUIMovie->Advance(deltaTime, 0);
    pRenderer->BeginFrame();

    if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
    {
        pRenderer->Display(hMovieDisplay);
    }

    pRenderer->EndFrame();
}
```

11 단계: 렌더링 상태 보호

```
// Save DirectX state before calling Scaleform
g_pStateBlock->Capture();
// Render the frame and advance the time counter
gfx->AdvanceAndRender();
// Restore DirectX state to avoid disturbing game render state
g_pStateBlock->Apply();
```

12 단계: 기본 UI 사용막기

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그림 3:

4.2 크기조절 모드

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```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```



```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

4.3 입력 이벤트 처리

4.3.1 마우스 이벤트

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                 bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);
    if (pUIMovie)
    {
        if (uMsg == WM_MOUSEMOVE)
        {
            MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONDOWN)
        {
            ::SetCapture(hWnd);
            MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
        else if (pMovieButton && uMsg == WM_LBUTTONUP)
        {
            ::ReleaseCapture();
            MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
        }
    }
}
```

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,  
                    screen_width, screen_height, 0);
```

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,  
                    screen_width / 4, screen_height / 4, 0);
```

```
movie_width = screen_width / 6;  
movie_height = screen_height / 6;  
pMovie->SetViewport( screen_width, screen_height,  
                    screen_width / 2 - movie_width / 2,  
                    screen_height / 2 - movie_height / 2,  
                    movie_width, movie_height);
```



4.3.2 키보드 이벤트

```
KeyEvent(EventType eventType = None,  
          Key::Code code = Key::None,  
          UByte asciiCode = 0,  
          UInt32 wcharCode = 0,  
          UInt8 keyboardIndex = 0)  
  
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

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```
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM  
lParam)
```

\

4.3.3 히트 테스트

```
bool processedMouseEvent = false;
if (uMsg == WM_MOUSEMOVE)
{
    MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONDOWN)
{
    ::SetCapture(hWnd);
    MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONUP)
{
    ::ReleaseCapture();
    MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}

if (processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
                                           Movie::HitTest_Shapes))
    *pbNoFurtherProcessing = true;
```

4.3.4 키보드 포커스

4.3.5 터치 이벤트

WINVER=0x601

```
TouchEvent( EventType evtType,  
            unsigned id,  
            float _x,  
            float _y,  
            float wcontact = 0,  
            float hcontact = 0,  
            bool primary = true,  
            float pressure = 1.0f)
```


5 지역화와 폰트 소개

5.1 폰트개요 및 기능

\\Data\\AS2\\ \\

早上好
你好!早上好!
多想你呀!
Chinese (繁體中文版)

그림 4a:

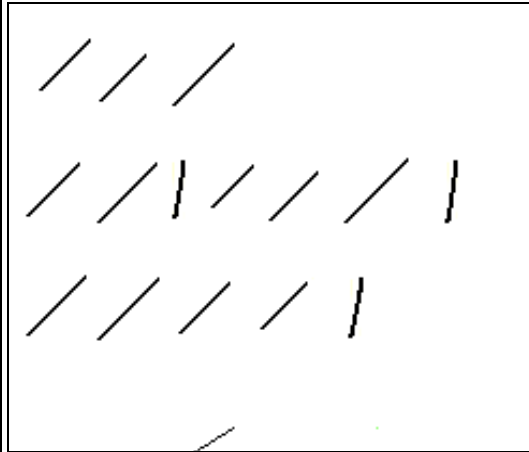


그림 4b:



그림 4c:

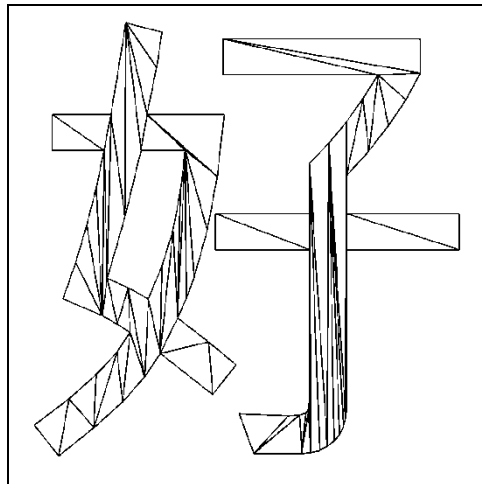


그림 4d:



그림 5:

6 C++, 플래시, 액션스크립트 연결하기

6.1 액션스크립트에서 C++로

6.1.1 FS Command 콜백

```
fscommand("setMode", "2");
```

\

```
class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                        const char* pcommand, const char* parg)
    {
        GFXPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

```
// Register our FSCommand handler
Ptr<GFXFSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

FSCommand: ToggleFullscreen, Args:

```
function toggleFullscreen(ev:MouseEvent) {  
    fscommand("ToggleFullscreen");  
}
```

```
if (strcmp(pcommand, "ToggleFullscreen") == 0)  
    doToggleFullscreen = true;
```

```
if (doToggleFullscreen)  
{  
    doToggleFullscreen = false;  
    DXUTToggleFullScreen();  
}
```

6.1.2 ExternalInterface

```

class OurExternalInterfaceHandler : public ExternalInterface
{
public:
virtual void Callback(Movie* pmovieView,
                      const char* methodName,
                      const Value* args,
                      unsigned argCount)
{
    GfxPrintf("ExternalInterface: %s, %d args: ",
              methodName, argCount);
    for(unsigned i = 0; i < argCount; i++)
    {
        switch(args[i].GetType())
        {
            case Value::VT_Null:
                GfxPrintf("NULL");
                break;
            case Value::VT_Boolean:
                GfxPrintf("%s", args[i].GetBool() ? "true" : "false");
                break;
            case Value::VT_Int:
                GfxPrintf("%s", args[i].GetInt());
                break;
            case Value::VT_Number:
                GfxPrintf("%3.3f", args[i].GetNumber());
                break;
            case Value::VT_String:
                GfxPrintf("%s", args[i].GetString());
                break;
            default:
                GfxPrintf("unknown");
                break;
        }
        GfxPrintf("%s", (i == argCount - 1) ? "" : ", ");
    }
    GfxPrintf("\n");
}
};

```

```

Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);

```

```

function onSubmit(path:String) {
    ExternalInterface.call("MeshPath", path);
    CloseMeshPath();
}

```

```

}

function onFocusIn(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", true);
}

function onFocusOut(ev: FocusHandlerEvent) {
    ExternalInterface.call("MeshPathFocus", false);
}

```

```

Callback! MeshPathFocus, nargs = 1
    arg(0) = true

```

```

Callback! MeshPathFocus, nargs = 1
    arg(0) = false

```

```

if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
args[0].GetType() == Value::VT_Boolean) {
    if (args[0].GetType() == Value::VT_Boolean)
        gfx->SetTextboxFocus(args[0].GetBool());
}

```

```

if (uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
    uMsg == WM_CHAR)
{
    if (textboxHasFocus || wParam == 32 || wParam == 9)
    {
        ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
        *pbNoFurtherProcessing = true;
    }
}

```



```

static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
    doChangeMesh = true;
    const char *filename = args[0].GetString();
    MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
        changeMeshFilename, _countof(changeMeshFilename));
}

```

6.2 C++에서 액션스크립트로

6.2.1 액션스크립트 변수 조작

\

```

void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,

```

```

                                LPARAM lParam, bool *pbNoFurtherProcessing)
{
    int mx = LOWORD(lParam), my = HIWORD(lParam);

    if (pHUDMovie && uMsg == WM_KEYDOWN)
    {
        if (wParam == VK_F5)
        {
            int counter = (int)pHUDMovie
                ->GetVariableDouble("_root.counter");
            counter++;
            pHUDMovie->SetVariable("_root.counter",
                Value((double)counter));

            char str[256];
            sprintf_s(str, "testing! counter = %d", counter);
            pHUDMovie->SetVariable("_root.MessageText.text", str);
        }
    }
    ...
}

```

성능관련 주의사항:



그림그림 8:

```
// Initialize the scene dropdown
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
```

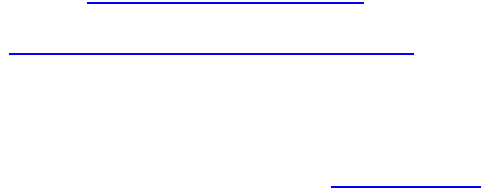
```
"_root.SceneData", 0, sceneData, 3);
```

```
\
```

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6.2.2 액션스크립트 서브루틴 실행

```
...  
else if (wParam == VK_F6)  
{  
    bool retval = pHUDMovie->Invoke("root.ui.hud.OpenMeshPath", "");  
    GfxPrintf("root.ui.hud.OpenMeshPath returns '%d'\n", (int) retval);  
}  
else if (wParam == VK_F7)  
{  
    const char *retval = pHUDMovie->Invoke(  
        "root.ui.hud.CloseMeshPath", "");  
    GfxPrintf("root.ui.hud.CloseMeshPath returns '%d'\n", (int) retval);  
}  
...
```



6.3 플래시 파일간 통신

```
var mcLoader:MovieClipLoader = new MovieClipLoader();
```

```
function loadMovie(url:String):void {  
    var loader = new Loader();  
    loader.load( new URLRequest( url ) );  
    addChild( loader )  
}
```

7 텍스처에 렌더링하기

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```

Material {
    1.000000;1.000000;1.000000;1.000000;;
    40.000000;
    1.000000;1.000000;1.000000;;
    0.000000;0.000000;0.000000;;
    TextureFilename {
    "cellwallRT.jpg";
    }
}

```

```

pd3dDevice->CreateTexture(rtw,rth,0,
    D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
    D3DP00L_DEFAULT, &g_pRenderTex, 0);

```

```

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();

```

a.

```

pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);

```

b.

```

ptex->GetSurfaceLevel(0, &psurface);
HRESULT hr = pd3dDevice->SetRenderTarget(0, psurface );

```

c.

8 OpenGL 샘플

9 GFxExport 로 전처리

```
gfxexport -i DDS -c d3d9guideAS3.swf  
gfxexport -i DDS -c fxplayer.swf
```

```
\          \          \  
\  

```

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10 다음 단계

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